

Conflict-Aware Cross-View Disaster Triage Across Spatial Observation Regimes [experiments]

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ABSTRACT

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage. Aggregate F1 scores, however, hide the cases where fusion matters most: examples for which the single-view predictors disagree. This study introduces a Conflict-Aware Evaluation protocol (CAE) for paired-view disaster triage. CAE reports overall F1, conflict-subset F1, and the conflict gain of cross-view fusion over the stronger single-view baseline. We apply CAE to two paired disaster settings: an Eaton/Altadena wildfire dataset with property-centered ground imagery and the Hurricane Ian CVDIsaster benchmark with panoramic ground imagery. The results show that cross-view fusion is most informative on disagreement cases, but the dominant single-view arbitrator depends on the observation regime and the split definition. On the wildfire sensitivity split, cross-view fusion improves conflict F1 by 0.0388 over the best single-view baseline. On a leakage-controlled hurricane endpoint split, cross-view fusion no longer dominates street-only inference on conflict cases, which provides an important boundary condition. We further report two pilot extensions: conflict density as an unsupervised tile-level wildfire damage proxy and a conflict-weighted loss that improves wildfire conflict F1. The evidence supports a conservative conclusion: paired-view disaster models should be evaluated as conflict resolvers, not only as average-case classifiers.

KEYWORDS

disaster assessment, cross-view learning, multimodal fusion, GeoAI, conflict-aware evaluation

1 INTRODUCTION

Post-disaster building assessment increasingly uses several visual viewpoints. Overhead imagery gives broad spatial coverage, while ground imagery can expose facade condition, debris, and local structural cues. Cross-view models seek to combine these signals, but prior evaluation often reduces their value to one aggregate test metric. That view is too coarse for operational disaster triage. Many buildings are easy in both views. The harder and more consequential cases are those where the two views provide competing evidence.

This paper studies cross-view fusion as a conflict-resolution problem. We focus on paired examples that contain a ground image, an overhead patch, and a binary damage label. Instead of asking only whether paired-view fusion improves average F1, we ask when fusion helps after the two single-view models disagree. That framing matters for geographic AI because observation conditions are not interchangeable. A close property inspection photograph and a panoramic street scene both count as ground imagery, but they

encode different relationships between observer, target building, and surrounding landscape.

The study sits between several lines of disaster-vision work. Satellite damage benchmarks such as xBD have made overhead-only building assessment a standard task [1], and later fusion models improve satellite-based damage assessment from multi-scale visual evidence [5]. Ground and street-view studies show that close-range imagery can capture damage cues that overhead imagery misses [6, 8, 9]. CVDIsaster is the closest paired-view benchmark for Hurricane Ian [3]. Recent generative work also studies how post-disaster ground views may be synthesized from satellite imagery when field observations are sparse [7]. CAE adds a complementary evaluation axis by separating average performance from behavior on single-view disagreement cases.

This study makes three contributions. First, it defines Conflict-Aware Evaluation (CAE), a three-metric protocol that reports full-test performance and conflict-subset behavior together. Second, it applies CAE to wildfire and hurricane paired-view triage, while treating the hurricane grouped-clean split as the conservative endpoint control. Third, it shows how the same disagreement signal can support two extensions: wildfire tile-level conflict-density mapping and a conflict-weighted training loss. The analysis deliberately avoids a causal claim that disaster type alone determines which view should dominate. The evidence is better read as a regime-sensitive experiment: cross-view fusion helps most when single-view evidence conflicts, and the source of that help depends on how the ground view observes the target.

2 DATA AND PROTOCOL

2.1 Paired-View Triage Settings

Let $\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$ be a paired disaster triage dataset, where $x_i^{(s)}$ is a street or ground image, $x_i^{(r)}$ is an overhead patch, and $y_i \in \{0, 1\}$ is the binary damage label. We compare three modes: *street-only*, *remote-only*, and *cross-view*. The cross-view model uses late fusion of ResNet18 features [2], including each embedding, their absolute difference, and their elementwise product. This simple fusion design keeps the evaluation focused on conflict behavior rather than architectural novelty.

The wildfire setting uses an Eaton/Altadena paired dataset assembled for this project. Its ground imagery is property-centered inspection imagery paired with overhead patches. This observation regime often places the target building near the visual center of the ground image. The hurricane setting uses the Hurricane Ian CVDIsaster benchmark [3], which pairs panoramic ground imagery with very high resolution overhead imagery. This observation regime often includes wider street context, adjacent buildings, and weaker

(a) Spatial Observation Regimes and Geographic Context

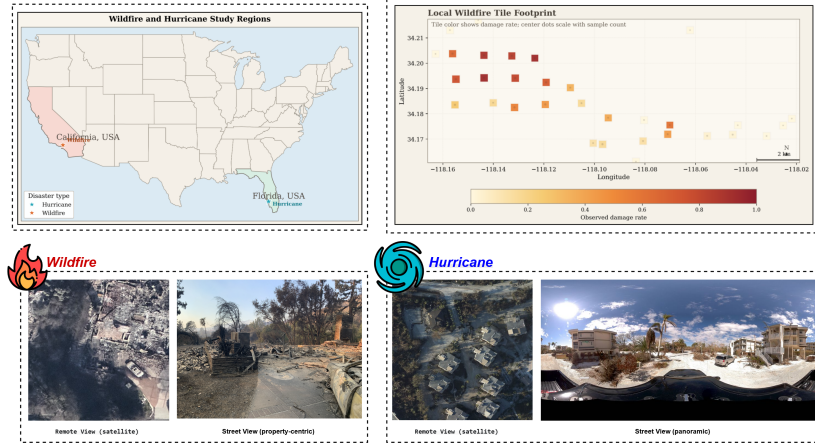


Figure 1: Study design and spatial observation regimes. The wildfire setting uses property-centered inspection imagery paired with overhead patches, whereas the hurricane setting uses panoramic ground imagery paired with overhead patches. This visual difference motivates the regime-aware interpretation tested by CAE.

Table 1: Evaluation settings and review controls.

Setting	Positive class	Ground regime	Role
WF sens.	affected+	property	main CAE
H end.	severe	panoramic	diagnostic
H mod.+sev.	moderate+	panoramic	sensitivity
H clean	severe	panoramic	leakage control

target alignment. The comparison therefore mixes disaster type, geography, source, and collection protocol; those factors limit causal interpretation and motivate the conservative language used below.

Table 1 records the experimental settings and the control role of each split. This is important because the paper should not choose a split after observing which one makes the conflict story look strongest. The wildfire sensitivity split is used because it gives a larger and more informative disagreement subset. The hurricane grouped-clean endpoint split is included because it controls object-level leakage in the original endpoint partition.

2.2 Conflict-Aware Evaluation

For each example, let $p_i^{(s)}$ and $p_i^{(r)}$ denote the positive-class probabilities predicted by the street-only and remote-only models. CAE defines the conflict subset

$$C_\tau = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}, \quad (1)$$

with $\tau = 0.1$ in the main analysis. For each model m , CAE reports standard overall F1 and conflict F1 on C_τ . It then computes the cross-view conflict gain

$$\Delta_{\text{view}} = \text{F1}(\text{crossview}; C_\tau) - \max\{\text{F1}(\text{street}; C_\tau), \text{F1}(\text{remote}; C_\tau)\}. \quad (2)$$

This protocol exposes whether paired-view fusion resolves disagreement cases rather than only improving easy average cases.

We also decompose conflicts by correction direction. $S \rightarrow O$ denotes cases where the street-only model is correct and the remote-only model is wrong; $O \rightarrow S$ denotes the reverse. These fractions

identify the stronger single-view arbitrator inside the conflict subset. Because a prior hurricane endpoint split had object-level leakage, we report the grouped-clean endpoint split as a required control rather than as an optional robustness run.

2.3 Implementation Details

All modes use the same triage implementation. Images are resized to 224 by 224 pixels and normalized for the selected backbone. The main runs use ImageNet pretrained ResNet18 encoders. Each view embedding is projected to 256 dimensions; the cross-view classifier receives the street embedding, overhead embedding, absolute difference, and elementwise product. Models are optimized with AdamW using learning rate 10^{-4} for 10 epochs, and the checkpoint with the best validation F1 is evaluated on the held-out test split. This keeps the comparison between street-only, remote-only, and cross-view models tied to the same training protocol.

3 RESULTS

3.1 CAE Summary

Table 2 reports the core CAE results. The wildfire sensitivity split has the clearest positive conflict gain: cross-view fusion reaches 0.6618 conflict F1, which is 0.0388 above the better single-view baseline. The original hurricane endpoint split also shows a large gain, but its small conflict subset ($n = 32$) and leakage history make it diagnostic rather than primary evidence. The broader hurricane moderate-plus-severe split keeps a smaller positive gain. The grouped-clean endpoint split provides the most important boundary condition: cross-view fusion remains close to street-only overall, but it no longer dominates street-only on the conflict subset.

The directional decomposition sharpens this interpretation. On wildfire, $S \rightarrow O$ conflicts are more common than $O \rightarrow S$ conflicts (0.2713 versus 0.1885), so the ground image is the stronger single-view arbitrator. On the hurricane endpoint split, the two directions are near tied (0.3750 versus 0.3438). On the broader hurricane split, the

Table 2: Conflict-Aware Evaluation summary. Conflict F1 uses C_τ with $\tau = 0.1$. WF is wildfire; H is hurricane. Gain is reported only for cross-view rows.

Setting	M	All	C-F1	Gain	n_c
WF sens.	S	0.9392	0.6230	–	435
WF sens.	R	0.9238	0.5237	–	435
WF sens.	CV	0.9473	0.6618	+0.0388	435
H end.	S	0.9055	0.6842	–	32
H end.	R	0.8969	0.5806	–	32
H end.	CV	0.9192	0.7647	+0.0805	32
H mod.+sev.	S	0.8738	0.7324	–	128
H mod.+sev.	R	0.8873	0.7742	–	128
H mod.+sev.	CV	0.8915	0.7867	+0.0125	128
H clean	S	0.9016	0.8101	–	80
H clean	R	0.8385	0.6526	–	80
H clean	CV	0.9008	0.7949	-0.0153	80

Table 3: Directional conflict decomposition at $\tau = 0.1$. $S \rightarrow O$ is street-correct and remote-wrong; $O \rightarrow S$ is the reverse.

Split	n_c	$S \rightarrow O$	$O \rightarrow S$	Stronger view
WF sens.	435	0.2713	0.1885	street
H end.	32	0.3750	0.3438	near-tie
H mod.+sev.	128	0.1094	0.1328	remote

ordering reverses (0.1094 versus 0.1328), making the overhead view the stronger single-view arbitrator. The stable claim is therefore not that one disaster always favors one modality. It is that conflict behavior changes with the observation regime and split definition.

3.2 Robustness Signals

The conflict effect is not only a single threshold artifact. Under the same $\tau = 0.1$ definition, cross-view accuracy on conflict cases is 0.7885 for the wildfire sensitivity split, compared with 0.7356 for street-only and 0.6529 for remote-only. On the hurricane moderate-plus-severe split, the corresponding values are 0.7500, 0.7031, and 0.7266. Label-independence permutation tests are below 10^{-4} in both settings. Multi-seed validation also supports the base ordering: wildfire cross-view reaches 0.9689 ± 0.0011 validation F1, while street-only and remote-only reach 0.9657 ± 0.0001 and 0.9654 ± 0.0015 . Hurricane cross-view reaches 0.9455 ± 0.0063 , compared with 0.9433 ± 0.0012 and 0.8959 ± 0.0065 . Backbone checks add a caution: ResNet18 and ResNet50 remain strong, while generic CLIP and DINOv2 features underperform the supervised CNN baselines in this disaster-specific triage setting.

3.3 Spatial and Training Extensions

CAE also produces useful secondary signals. For the wildfire dataset, we aggregate $|p_i^{(s)} - p_i^{(r)}|$ by overhead tile and compare the resulting conflict density with observed tile damage rate. Across 27 tiles, Spearman $r = 0.5123$ and $p = 0.0063$. This suggests that cross-view disagreement can act as an unsupervised spatial damage proxy in the property-centered wildfire setting. We do not yet claim the same result for hurricane because the benchmark does not provide equivalent tile metadata in this repository.

We also test ConflictFocalLoss, a conflict-weighted binary loss related in spirit to focal reweighting [4]. For each cross-view example, the loss multiplies binary cross entropy by $(1 + c_i)$, where

Table 4: Pilot method and spatial extensions.

Extension	Primary statistic	Scope
Conflict density	$r = 0.5123, p = 0.0063$	wildfire tiles
ConflictFocalLoss	conflict F1 = 0.7208	wildfire split
Adaptive cascade	F1 = 0.9498, 9.5% routed	wildfire split

c_i is a feature-space disagreement score based on the cosine distance between the two view embeddings. On the wildfire sensitivity split, moderate conflict weighting improves overall F1 from 0.9473 to 0.9520 and conflict F1 from 0.6618 to 0.7208. This result is encouraging, but it is still a single-setting pilot rather than a fully validated method contribution.

4 DISCUSSION AND LIMITATIONS

The experiment supports CAE as a useful evaluation protocol for paired-view disaster triage. Aggregate F1 alone would make the models look similar in some settings. CAE reveals how they behave on disagreement cases and shows that cross-view value can shrink, grow, or even become negative relative to the stronger single-view model after leakage control.

4.1 Reproducibility and Boundary Conditions

The study should be read as an evaluation protocol plus an empirical comparison, not as a final claim that one universal fusion architecture is optimal. The reproducible unit is the CAE computation: given held-out predictions from street-only, remote-only, and cross-view models, CAE can be computed without changing the model architecture. This makes the protocol portable to other paired-view disaster datasets and to stronger fusion models, including attention-based, transformer-based, and generative-view designs.

Two controls are especially important for credible reuse. First, the hurricane endpoint result must be interpreted through the grouped-clean split. The original endpoint split is still useful because it connects to the benchmark history, but it should not carry the main conclusion after object-level leakage has been identified. Second, the wildfire dataset needs a complete data card: sample counts by split, class balance, image source, collection timing, annotation rules, quality-control procedure, and release or access constraints. Without that documentation, external users cannot separate a modeling result from a hidden data-construction effect.

The spatial-map result has a similar boundary. The tile-level correlation is useful because it turns disagreement into a GIS-native damage signal, but the current statistic uses only 27 wildfire tiles. A final version should either report a spatial-autocorrelation check or explicitly describe the Spearman result as an exploratory spatial association. The same conservative stance applies to ConflictFocalLoss. Its wildfire gain is promising, but the loss should be compared against standard focal loss, class-balanced loss, and simple probability averaging before being framed as a standalone method contribution.

Several limits are central. First, the wildfire and hurricane settings differ by more than observation regime. They also differ by hazard, geography, image source, label construction, and metadata

availability. The results therefore support a regime-sensitive interpretation, not a causal proof. Second, the hurricane endpoint conflict subset is small, so its large positive gain should be treated as tentative. Third, the private wildfire data package must include complete counts, class balance, annotation protocol, inter-annotator agreement if available, and release or access instructions before public release. Fourth, ConflictFocalLoss and the tile-level spatial proxy are validated only on the wildfire setting. They motivate follow-up experiments but should not be presented as fully general.

These limits also clarify the paper's contribution. CAE does not claim that cross-view fusion always wins. It gives researchers and practitioners a clearer way to see when paired views resolve evidence conflict and when a single view remains sufficient.

4.2 Relation to Generative Cross-View Assessment

The results also identify a practical evaluation target for generative satellite-to-street systems. When post-disaster street imagery is missing, a model may synthesize a plausible ground-level view from overhead imagery [7]. Such generated evidence should not be evaluated only by perceptual realism. In disaster triage, the operational question is whether a synthetic or retrieved ground view changes the decision on cases where overhead evidence is ambiguous. CAE provides a direct test: generated, real, and overhead-only predictions can be compared on the same conflict subset, and the result can show whether the added view resolves uncertainty or merely reinforces the overhead model's prior.

This distinction matters for responsible GeoAI. A visually realistic generated street scene can still hallucinate facade details, debris patterns, or access conditions that are decisive for damage labels. Conversely, a generated view that is imperfect as an image may still add useful structural evidence if its errors are calibrated and its uncertainty is surfaced. Conflict-aware reporting therefore complements view synthesis by asking how synthetic evidence behaves under decision conflict, where the cost of a plausible but wrong cue is highest.

4.3 Operational Implications

The practical use of CAE is a routing view of disaster triage. If street-only and remote-only predictions agree with high confidence, a response system can often avoid the cost of paired-view inference and use the cheaper agreement decision. If the two predictions disagree, the case becomes a candidate for cross-view fusion or human review. This is the logic behind the adaptive cascade result in Table 4: only 9.5% of wildfire samples are routed to the cross-view stage while preserving the cross-view performance level. The exact routing threshold should be tuned for each deployment, but the principle is simple: cross-view reasoning is most valuable when the evidence is already in tension.

The same view can guide dataset design. If a future benchmark wants to test observation-regime effects directly, it should collect property-centered and panoramic ground imagery for the same disaster and the same target buildings. That design would remove many of the confounders in the current comparison and would let CAE test whether target alignment, rather than hazard type or data

source, drives the dominant-view switch. In this sense, the study offers an empirical warning and an evaluation protocol: average F1 is not enough for multimodal disaster assessment when the views disagree.

CAE can also support audit logs for triage escalation: the two single-view scores, conflict threshold, selected route, and final decision can be stored for each building. These records help target new ground photos where overhead imagery is uncertain and prior conflicts show that ground evidence often corrects remote-only mistakes.

5 CONCLUSION

This work reframes paired-view disaster triage as conflict-aware evaluation. Across wildfire and hurricane paired-view settings, cross-view fusion is most informative on disagreement cases, but the stronger single-view arbitrator changes with observation regime, split definition, and leakage control. The resulting lesson is practical: disaster benchmarks should report conflict-subset behavior alongside aggregate scores. Doing so makes multimodal fusion more falsifiable, more spatially interpretable, and more useful for deployment decisions.

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